

What are the main three types of sum of squares in al linear regression? Define each one conceptually.

The three main types of sum of squares in linear regression are **Total Sum of Squares (SST)**, **Regression Sum of Squares (SSR)**, and **Residual Sum of Squares (SSE)**. SST measures the total variation in the dependent variable, SSR measures the variation explained by the model, and SSE measures the variation that is not explained by the model.

1. Total Sum of Squares (SST)

- This represents the total variability in the dependent variable, calculated as the sum of the squared differences between each data point and the mean of the dependent variable.

Formula: $SST = \sum (y_i - \bar{y})^2$.

2. Regression Sum of Squares (SSR)

- Also known as the explained sum of squares (ESS), this measures the variability in the dependent variable that is explained by the regression model.
- It is the sum of the squared differences between the predicted values and the mean of the dependent variable.

Formula: $SSR = \sum (\hat{y}_i - \bar{y})^2$.

3. Residual Sum of Squares (SSE)

- Also known as the error sum of squares, this measures the unexplained variation in the dependent variable.
- It is the sum of the squared differences between the actual values and the predicted values from the regression model.

Formula: $SSE = \sum (y_i - \hat{y}_i)^2$.

The relationship between these sums of squares is expressed by the equation:

$$SST = SSR + SSE$$

This shows that the total variation in the data is the sum of the variation explained by the model and the variation that the model fails to explain.